Questions.

Question 1 [20 marks]

a) Briefly explain the following terms.
   i. Race conditions
   ii. Starvation
   iii. Critical section

[6 Marks]

b) What is meant by "Process Synchronization" and why is it important?

[5 Marks]

c) Consider the following methods used by the process "A" and "B" accessing their critical section as given bellow. The elements of the shared boolean array **"entry"** will be initialized by value "False". Does this algorithm avoid the critical section problem? Justify your answer by clearly explaining each step.

```
boolean entry[2];  //Define as a global variable
entry [0] = FALSE;   entry [1] = FALSE // Set Initial values
```

A	B

`entry[0] = TRUE;`
`while (entry[1]);`
`critical section`
`entry[0] = FALSE;`
 	
`remainder section`
	`entry[1] = TRUE;`
	`while (entry[0]);`
	`critical section`
	`entry[1] = FALSE;`
	
	`remainder section`

[9 Marks]

Question 2 [20 marks]

a) Four jobs are executed in a single processer system and their arrival-time to ready queue, CPU burst-time and I/O burst-time are given in the following table. **Shortest Job First** (SJF) with **preemption** scheduling algorithm is used as the CPU scheduler. For I/O operations, jobs are move to I/O waiting queue and I/O queue uses First Come First Serve scheduling algorithm to cater I/O operations.

Process ID	**Arrival Time**	**Processing Time (CPU Burst, I/O Burst , CPU Burst)**
P1	0	(3,3,1)
P2	0	(1,3,1)
P3	6	(6,0,0)
P4	7	(2,3,1)

i. Create a Gantt chart to illustrate the execution of the above processes in both queues.
ii. Calculate the average waiting time and the average turnaround time of the above processes.

Note: Clearly mention the computation steps and any assumption you made.

[13 Marks]

b) A company is installed a shared printer to get printout of their multiple users effectively. Among the main scheduling algorithms such as "First Come First Serve", "Shortest-Job-First", "Shortest-Remaining-Job-First", "Priority" and "Round Robin", which algorithm(s) is (are) not appropriate to manage the above printer queue? Justify your answer. Propose an algorithm (standard algorithm or you can propose the new algorithm) to gain the best throughput in the above scenario.

[7 Marks]

Question 03 [15 marks]

a) Consider the following table that includes maximum utilization and current allocation of resources (R0, R1, R2 and R3) among the five (05) processes (P0, P1, P2, P3 and P4). Resources R0, R1, R2 and R3 have instances 3, 14, 12 and 12, respectively.

Is this system can run the processes without moving to a deadlock? If possible provide the safe execution sequence. Justify your answer.

Processes	Allocated Resources				Maximum Utilization			
	R0	R1	R2	R3	R0	R1	R2	R3
P0	1	0	1	0	1	7	5	0
P1	1	2	5	4	2	3	5	6
P2	0	1	1	4	0	6	5	6
P3	0	1	0	2	0	1	1	2
P4	0	6	3	2	0	6	5	2

[10 marks]

b) Figure 1 shows resources allocation graph of a system. Is it in a safe state? Justify your answer.

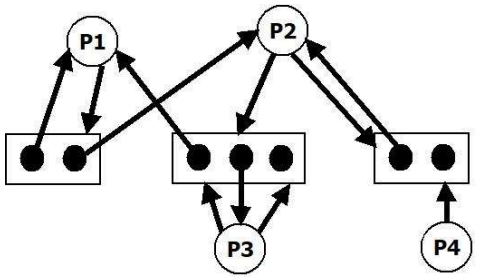


Figure 1

[5 marks]

Question 04 **[25 marks]**

a) What is a page fault with regards to the memory access? Explain how an operating system accommodates a page fault when it occurred.

[5 marks]

b) Virtual memory system is running demand paging algorithms with three frames. Consider the following virtual page reference string
1, 2, 1, 3, 7, 4, 5, 6, 3, 1
At the initially stage, page 2 and 4 are in the frame memory. How many page faults would occur for the Least Recently Used (LRU) replacement algorithm? Clearly mention the computation steps and any assumption you made.

[10 marks]

c) A system with 64 logical addresses space uses a two-level page algorithm, in which the page table itself is also paged (figure 2). Assume that the word size is one byte and clearly mention the computation steps.

outer page	*inner page*	*Page offset*
32	18	14

Figure 2

i. What is the frame size?
ii. What is the size of the inner page table?
iii. What is the maximum number of inner page tables in this system?

[10 marks]